Supplementary Files for "A Clustering Aided Multi-Operator Differential Evolution via Cluster-Level Adaptation"

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Table S1: Parameter settings for CEC competitions

Algorithm	Parameter setting
CAMODE-CLA	$NP = 6 \times D^2$, $NP_{min} = 4$, $K_{init} = \text{round}(NP/10)$, $\gamma \in [0.3, 0.7]$, $\alpha = 0.15$.
IMODE	$NP = 6 \times D^2$, $NP_{min} = 4$, $Prob_{ls} = 0.1$.
MadDE	$NP = 6 \times D^2$, $NP_{min} = 4$, $p = 0.18$, $H = 10 \times D$, $\gamma = 0.1$.
APGSK-IMODE	$NP = 6 \times D^2$, $PS_2 = \text{round}(NP/4)$, $PS_1 = NP - PS_2$, $NP_{1,min} = 4$, $CS = 100$.
EA4eig	$NP = 100$, $NP_{min} = 10$, $P_{eig} = 0.4$, $CB_{ps} = 0.5$.
mLSHADE-RL	$NP = 6 \times D^2$, $NP_{min} = 4$, $r^{arc} = 1.4$, $H = 20 \times D$, $P_b = 0.4$, $n_opr = 3$.
UMOEAs-III	$NP = 6 \times D^2$, $NP_{min} = 8$, $prob_{ls} = 0.1$, $n_{ops} = 2$, $H = 20 \times D$, $PS_{1,min} = 8$, $PS2 = 4 + \lfloor (3 \log(D)) \rfloor$, $CS = 50$ for 5D functions, $CS = 100$ for 10D functions, $CS = 150$ for 15D functions, $CS = 200$ for 20D functions.

Table S2: Details of Benchmark Suites

Suite	Funcs.	Dims (D) / MAX_FES	Problem Characteristics
CEC2020	10	5D (5×10^4) 10D (1×10^6) 15D (3×10^6) 20D (1×10^7)	F_1: Unimodal. $F_2 - F_4$: Basic functions. $F_5 - F_7$: Hybrid functions. $F_8 - F_{10}$: Composition functions.
CEC2021	10	10D (1×10^6) 20D (1×10^7)	Bias, shift, and rotation
CEC2022	12	10D (1×10^6) 20D (1×10^7)	Complex hybrid compositions

Table S3: Parameter settings for Real World Problems

Algorithm	Parameter setting
CAMODE-CLA	$NP = 75$, $NP_{min} = 4$, $K_{init} = \text{round}(NP/10)$, $\gamma \in [0.3, 0.7]$, $\alpha = 0.15$.
IMODE	$NP = 60$, $NP_{min} = 4$, $Prob_{ls} = 0.1$.
MadDE	$NP = 70$, $NP_{min} = 4$, $p = 0.18$, $H = 5 \times D$, $\gamma = 0.1$.
APGSK-IMODE	$NP = 70$, $PS_2 = \text{round}(NP/4)$, $PS_1 = NP - PS_2$, $NP_{1,min} = 4$, $CS = 50$.
EA4eig	$NP = 40$, $NP_{min} = 5$, $P_{eig} = 0.4$, $CB_{ps} = 0.5$.
mLSHADE-RL	$NP = 60$, $NP_{min} = 4$, $r^{arc} = 1.4$, $H = 5 \times D$, $P_b = 0.4$, $n_opr = 3$.
UMOEAs-III	$NP = 30$, $NP_{min} = 4$, $prob_{ls} = 0.1$, $n_{ops} = 2$, $H = 20 \times D$, $PS_{1,min} = 8$, $PS2 = 4 + \lfloor (3 \log(D)) \rfloor$, $CS = 100$

Table S4: Experimental results of CAMODE-CLA and comparison algorithms for the 5D and 10D CEC2020 test set.

Func		APGSK-IMODE		EA4eig		IMODE		MadDE		UMOEAs-III		mLSHADE-RL		CAMODE-CLA
							5D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	**1.03e-01**	=	4.15e+00	+	1.50e-01	=	6.48e-01	+	4.66e+00	+	3.56e-01	=	3.28e-01
	Std	**1.03e-01**		2.17e+01		1.01e-01		1.67e+00		2.16e+01		1.25e+00		1.21e+00
F3	Mean	4.81e+00	=	4.50e+00	=	5.16e+00	=	5.44e+00	+	5.20e+00	=	5.24e+00	+	**4.50e+00**
	Std	1.09e+00		1.80e+00		4.29e-02		2.33e-01		1.16e-01		1.10e-01		**1.59e+00**
F4	Mean	**0.00e+00**	=	8.45e-02	+	**0.00e+00**	=	**0.00e+00**	=	6.16e-02	=	9.15e-02	+	**0.00e+00**
	Std	**0.00e+00**		4.25e-02		**0.00e+00**		**0.00e+00**		5.36e-02		2.72e-02		**0.00e+00**
F5	Mean	**0.00e+00**	=	8.71e-02	=	**0.00e+00**	=	**0.00e+00**	=	2.62e-01	+	3.32e-02	=	**0.00e+00**
	Std	**0.00e+00**		3.42e-01		**0.00e+00**		**0.00e+00**		3.92e-01		1.82e-01		**0.00e+00**
F6	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	2.08e-02	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		1.14e-01		**0.00e+00**		**0.00e+00**
F7	Mean	**0.00e+00**	=	3.95e+00	=	**0.00e+00**	=	**0.00e+00**	=	1.99e+01	+	7.91e+00	=	**0.00e+00**
	Std	**0.00e+00**		2.16e+01		**0.00e+00**		**0.00e+00**		4.52e+01		3.00e+01		**0.00e+00**
F8	Mean	**0.00e+00**	=	3.33e-01	=	**0.00e+00**	=	**0.00e+00**	=	6.67e+00	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		1.82e+00		**0.00e+00**		**0.00e+00**		2.54e+01		**0.00e+00**		**0.00e+00**
F9	Mean	5.00e+01	+	9.00e+01	+	**0.00e+00**	=	2.33e+01	=	9.00e+01	+	9.33e+01	+	3.28e+00
	Std	5.09e+01		3.05e+01		**0.00e+00**		4.30e+01		5.48e+01		2.54e+01		1.25e+01
F10	Mean	2.46e+02	=	3.43e+02	+	3.08e+02	+	3.08e+02	+	3.44e+02	+	3.47e+02	+	**2.01e+02**
	Std	1.25e+02		1.45e+01		9.50e+01		8.57e+01		1.20e+01		0.00e+00		**1.17e+02**
	+/=/-	1+/9=/0-		4+/6=/0-		1+/9=/0-		3+/7=/0-		6+/4=/0-		4+/6=/0-		
							10D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	3.45e+00	=	5.52e+00	=	3.07e+00	-	2.67e+00	-	1.96e+00	-	**9.29e-01**	-	5.37e+00
	Std	3.03e+00		4.54e+00		2.33e+00		2.56e+00		2.97e+00		**1.32e+00**		4.08e+00
F3	Mean	1.22e+01	+	1.05e+01	=	1.24e+01	+	1.21e+01	+	1.05e+01	=	1.11e+01	+	**9.83e+00**
	Std	6.40e-01		1.78e+00		6.49e-01		5.50e-01		1.78e-01		4.20e-01		**2.67e+00**
F4	Mean	**0.00e+00**	=	2.22e-01	+	**0.00e+00**	=	**0.00e+00**	=	1.56e-01	+	8.76e-02	+	**0.00e+00**
	Std	**0.00e+00**		4.92e-02		**0.00e+00**		**0.00e+00**		3.27e-02		8.30e-02		**0.00e+00**
F5	Mean	7.73e-01	+	6.24e-02	-	4.93e-01	+	**2.78e-02**	-	8.37e+00	+	2.17e+01	+	2.08e-01
	Std	6.08e-01		1.11e-01		4.94e-01		**7.20e-02**		2.35e+01		3.58e+01		1.55e-01
F6	Mean	4.95e-02	-	**3.67e-02**	-	1.18e-01	=	1.46e-01	+	2.72e-01	+	7.31e-02	-	1.06e-01
	Std	2.46e-02		**8.44e-02**		7.32e-02		9.01e-02		2.20e-01		6.13e-02		8.01e-02
F7	Mean	1.74e-03	=	1.04e-02	+	**1.09e-03**	=	1.14e-02	=	3.27e-01	+	1.01e-01	=	1.33e-02
	Std	2.30e-03		5.70e-02		**1.15e-03**		4.82e-02		2.47e-01		1.65e-01		5.67e-02
F8	Mean	2.60e+01	=	9.67e+01	+	**2.41e+00**	-	6.79e+01	+	1.00e+02	+	1.00e+02	+	8.45e+00
	Std	4.19e+01		1.83e+01		**7.68e+00**		3.72e+01		0.00e+00		0.00e+00		9.25e+00
F9	Mean	7.67e+01	=	2.53e+02	+	**4.29e+01**	=	9.00e+01	+	1.60e+02	+	1.51e+02	+	5.30e+01
	Std	4.30e+01		1.03e+02		**4.55e+01**		3.05e+01		9.75e+01		9.40e+01		4.85e+01
F10	Mean	2.69e+02	=	4.09e+02	+	3.88e+02	+	3.98e+02	+	4.10e+02	+	4.06e+02	+	**1.43e+02**
	Std	1.50e+02		1.95e+01		5.44e+01		0.00e+00		2.03e+01		1.71e+01		**1.03e+02**
	+/=/-	2+/7=/1-		5+/3=/1-		3+/5=/2-		5+/3=/2-		7+/2=/1-		5+/3=/2-		

Table S5: Experimental results of CAMODE-CLA and comparison algorithms for the 15D and 20D CEC2020 test set.

Func		APGSK-IMODE		EA4eig		IMODE		MadDE		UMOEAs-III		mLSHADE-RL		CAMODE-CLA
							15D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	4.46e+00	=	3.18e+01	+	3.19e+00	=	**6.07e-01**	-	3.71e+01	+	6.06e+00	+	3.99e+00
	Std	4.58e+00		4.57e+01		2.89e+00		**1.34e+00**		5.17e+01		2.13e+01		3.51e+00
F3	Mean	1.61e+01	+	1.65e+01	+	1.61e+01	+	1.56e+01	+	1.58e+01	+	1.66e+01	+	**1.06e+01**
	Std	3.41e-01		6.76e-01		3.96e-01		5.69e-02		2.10e-01		4.24e-01		**7.18e+00**
F4	Mean	**0.00e+00**	=	3.60e-01	+	**0.00e+00**	=	**0.00e+00**	=	2.42e-01	+	6.10e-02	+	**0.00e+00**
	Std	**0.00e+00**		7.05e-02		**0.00e+00**		**0.00e+00**		6.95e-02		1.05e-01		**0.00e+00**
F5	Mean	8.46e+00	=	**1.95e+00**	-	6.93e+00	-	2.61e+00	-	8.58e+00	-	1.44e+02	+	9.25e+00
	Std	4.35e+00		**2.11e+00**		2.58e+00		1.09e+00		3.03e+01		9.37e+00		3.75e+00
F6	Mean	3.47e-01	-	**1.54e-01**	-	6.56e-01	=	2.62e-01	-	7.29e-01	=	3.97e-01	-	6.45e-01
	Std	1.53e-01		**1.08e-01**		2.44e-01		1.66e-01		1.36e-01		1.79e-01		2.63e-01
F7	Mean	3.63e-01	-	**2.91e-01**	+	6.09e-01	=	5.33e-01	=	5.81e-01	=	2.14e+01	+	5.26e-01
	Std	1.66e-01		**2.33e-01**		2.37e-01		1.09e-01		1.04e-01		4.45e+01		1.91e-01
F8	Mean	4.66e+01	+	1.00e+02	+	**2.06e+00**	=	1.00e+02	+	1.00e+02	+	1.00e+02	+	6.09e+00
	Std	4.63e+01		0.00e+00		**6.57e+00**		0.00e+00		0.00e+00		0.00e+00		1.01e+01
F9	Mean	9.67e+01	=	3.74e+02	+	**9.00e+01**	=	1.00e+02	=	3.10e+02	+	3.64e+02	+	1.00e+02
	Std	1.83e+01		3.38e+01		**3.05e+01**		0.00e+00		1.27e+02		5.34e+01		0.00e+00
F10	Mean	4.00e+02	+	4.00e+02	+	4.00e+02	+	4.00e+02	+	4.00e+02	+	4.00e+02	+	**3.10e+02**
	Std	0.00e+00		0.00e+00		0.00e+00		0.00e+00		0.00e+00		0.00e+00		**1.40e+02**
	+/=/-	3+/5=/2-		7+/1=/2-		2+/7=/1-		3+/4=/3-		6+/3=/1-		8+/1=/1-		
							20D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	6.66e-01	=	4.99e+00	+	4.67e-01	=	**5.00e-02**	-	7.83e+01	+	1.22e-01	-	6.15e-01
	Std	8.04e-01		3.66e+00		6.61e-01		**3.90e-02**		1.18e+02		4.52e-02		7.66e-01
F3	Mean	2.04e+01	+	2.25e+01	+	2.05e+01	+	2.04e+01	+	2.05e+01	+	2.16e+01	+	**1.63e+01**
	Std	1.01e-01		1.15e+00		1.33e-01		0.00e+00		1.40e-01		3.99e-01		**8.23e+00**
F4	Mean	**0.00e+00**	=	4.97e-01	+	**0.00e+00**	=	**0.00e+00**	=	3.04e-01	+	1.52e-02	+	**0.00e+00**
	Std	**0.00e+00**		1.18e-01		**0.00e+00**		**0.00e+00**		4.69e-02		4.48e-02		**0.00e+00**
F5	Mean	9.16e+00	+	3.18e+01	=	1.15e+01	+	3.21e+00	=	4.30e+00	-	**1.70e+00**	-	4.65e+00
	Std	4.13e+00		5.17e+01		5.41e+00		1.45e+00		2.16e+01		**2.05e+00**		3.14e+00
F6	Mean	2.02e-01	=	1.52e-01	-	2.88e-01	+	**1.42e-01**	-	7.70e-01	+	1.17e+00	+	1.97e-01
	Std	7.86e-02		6.86e-02		8.67e-02		**4.65e-02**		1.23e-01		5.14e-02		8.32e-02
F7	Mean	**3.25e-01**	-	8.98e-01	-	5.43e-01	-	4.85e-01	=	5.34e+00	+	6.92e+01	+	9.89e-01
	Std	**1.34e-01**		2.11e+00		1.61e-01		2.12e-01		2.17e+01		5.79e+01		2.12e+00
F8	Mean	9.28e+01	+	1.00e+02	+	8.22e+01	+	1.00e+02	+	1.00e+02	+	1.00e+02	+	**4.75e+01**
	Std	1.39e+01		0.00e+00		2.17e+01		0.00e+00		0.00e+00		0.00e+00		**2.29e+01**
F9	Mean	1.04e+02	=	4.05e+02	+	**8.62e+01**	-	1.41e+02	+	3.80e+02	+	3.89e+02	+	1.00e+02
	Std	2.12e+01		4.49e+00		**3.19e+01**		1.06e+02		6.10e+01		5.85e+00		0.00e+00
F10	Mean	**3.99e+02**	=	4.06e+02	+	4.00e+02	+	4.14e+02	+	4.14e+02	+	4.14e+02	+	**3.99e+02**
	Std	**1.17e-02**		1.06e-02		2.59e-01		0.00e+00		6.35e-01		2.57e-02		**3.77e-01**
	+/=/-	3+/6=/1-		6+/2=/2-		5+/3=/2-		4+/4=/2-		8+/1=/1-		7+/1=/1-		

Table S6: Experimental results of CAMODE-CLA and comparison algorithms for the 10D and 20D CEC2021 test set.

Func		APGSK-IMODE		EA4eig		IMODE		MadDE		UMOEAs-III		mLSHADE-RL		CAMODE-CLA
							10D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	3.74e+00	=	7.23e+00	=	3.68e+00	-	3.48e+00	-	2.80e+00	-	**4.96e-01**	-	5.97e+00
	Std	2.53e+00		8.54e+00		2.87e+00		2.54e+00		3.45e+00		**8.40e-01**		3.62e+00
F3	Mean	1.19e+01	+	1.08e+01	=	1.22e+01	+	1.25e+01	+	1.05e+01	+	1.13e+01	+	**9.80e+00**
	Std	6.34e-01		3.82e-01		7.36e-01		5.00e-01		1.27e-01		3.64e-01		**2.80e+00**
F4	Mean	9.45e-02	+	3.39e-01	+	**3.29e-03**	-	2.45e-01	+	1.61e-01	+	3.18e-01	+	3.56e-02
	Std	6.59e-02		7.75e-02		**4.73e-03**		8.87e-02		4.37e-02		4.85e-02		2.99e-02
F5	Mean	9.39e-01	+	5.54e+00	+	5.01e-01	+	**9.71e-02**	-	6.32e+00	+	2.34e+01	+	2.34e-01
	Std	6.18e-01		2.98e+01		4.24e-01		**1.31e-01**		1.02e+01		4.00e+01		2.36e-01
F6	Mean	9.50e-02	=	**1.85e-02**	-	1.08e-01	=	1.72e-01	+	2.49e-01	+	6.47e-02	=	9.23e-02
	Std	7.55e-02		**2.63e-02**		8.15e-02		1.10e-01		1.59e-01		3.95e-02		6.70e-02
F7	Mean	2.78e-03	-	1.04e-02	-	**7.80e-04**	-	2.97e-03	-	2.48e-01	+	6.68e-02	=	6.75e-02
	Std	3.75e-03		5.70e-02		**1.01e-03**		6.91e-03		2.16e-01		1.29e-01		1.50e-01
F8	Mean	1.91e+01	=	9.04e+01	+	**9.09e-01**	-	7.92e+01	+	1.00e+02	+	1.00e+02	+	6.60e+00
	Std	3.38e+01		2.94e+01		**4.98e+00**		3.02e+01		0.00e+00		0.00e+00		8.17e+00
F9	Mean	8.11e+01	=	2.59e+02	+	**5.97e+01**	=	9.00e+01	+	1.63e+02	+	1.48e+02	+	6.99e+01
	Std	3.89e+01		9.84e+01		**4.79e+01**		3.05e+01		9.75e+01		9.87e+01		4.43e+01
F10	Mean	3.08e+02	+	4.13e+02	+	3.68e+02	+	3.98e+02	+	4.09e+02	+	4.07e+02	+	**1.30e+02**
	Std	1.39e+02		2.20e+01		9.08e+01		0.00e+00		1.95e+01		1.85e+01		**9.08e+01**
	+/=/-	4+/4=/2-		5+/3=/2-		3+/3=/4-		6+/1=/3-		8+/1=/1-		6+/3=/1-		
							20D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	2.35e-01	=	4.66e+00	+	4.58e-01	=	**4.68e-02**	-	4.72e+01	+	1.39e-01	=	4.60e-01
	Std	4.11e-01		3.35e+00		6.59e-01		**3.91e-02**		8.80e+01		4.97e-02		6.71e-01
F3	Mean	2.04e+01	+	2.20e+01	+	2.05e+01	+	2.04e+01	+	2.05e+01	+	2.16e+01	+	**1.77e+01**
	Std	8.09e-02		8.66e-01		1.37e-01		0.00e+00		1.44e-01		4.93e-01		**6.88e+00**
F4	Mean	5.19e-01	+	7.41e-01	+	4.74e-01	=	4.24e-01	=	**3.40e-01**	-	5.56e-01	+	4.42e-01
	Std	7.01e-02		1.44e-01		7.66e-02		5.66e-02		**5.03e-02**		5.97e-02		1.01e-01
F5	Mean	7.95e+00	+	1.30e+01	=	1.04e+01	+	**3.45e+00**	=	8.21e+00	+	2.41e+02	+	4.20e+00
	Std	3.72e+00		3.06e+01		5.11e+00		**1.99e+00**		3.01e+01		7.08e+01		3.33e+00
F6	Mean	2.17e-01	=	1.95e-01	=	2.73e-01	+	**1.25e-01**	-	7.34e-01	+	1.16e+00	+	2.10e-01
	Std	8.30e-02		1.03e-01		8.63e-02		**5.50e-02**		1.43e-01		5.46e-02		9.17e-02
F7	Mean	**3.46e-01**	=	6.00e-01	-	4.83e-01	=	4.35e-01	=	8.76e-01	+	4.17e+01	+	6.97e-01
	Std	**1.31e-01**		1.51e+00		1.66e-01		2.09e-01		6.69e-01		5.11e+01		1.58e+00
F8	Mean	9.36e+01	+	1.00e+02	+	8.68e+01	+	1.00e+02	+	1.00e+02	+	1.00e+02	+	**3.83e+01**
	Std	1.32e+01		0.00e+00		1.91e+01		0.00e+00		0.00e+00		0.00e+00		**2.63e+01**
F9	Mean	1.00e+02	=	4.04e+02	+	**9.18e+01**	=	2.09e+02	+	3.83e+02	+	3.88e+02	+	9.33e+01
	Std	0.00e+00		4.28e+00		**2.64e+01**		1.53e+02		6.47e+01		7.67e+00		2.54e+01
F10	Mean	**3.99e+02**	=	4.06e+02	+	4.00e+02	=	4.14e+02	+	4.14e+02	+	4.14e+02	+	**3.99e+02**
	Std	**2.04e-02**		6.91e-01		3.55e-01		0.00e+00		6.46e-01		1.32e-02		**3.90e-01**
	+/=/-	4+/6=/0-		6+/3=/1-		4+/6=/0-		4+/4=/2-		8+/1=/1-		8+/2=/0-		

Table S7: Experimental results of CAMODE-CLA and comparison algorithms for the 10D and 20D CEC2022 test set.

Func		APGSK-IMODE		EA4eig		IMODE		MadDE		UMOEAs-III		mLSHADE-RL		CAMODE-CLA
							10D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	**0.00e+00**	=	3.99e+00	+	**0.00e+00**	=	**0.00e+00**	=	2.78e+00	+	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		1.95e+00		**0.00e+00**		**0.00e+00**		3.09e+00		**0.00e+00**		**0.00e+00**
F3	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F4	Mean	3.45e+00	-	2.98e+00	-	7.20e+00	-	5.07e+00	-	**0.00e+00**	-	8.62e-01	-	9.52e+00
	Std	8.15e-01		8.15e-01		1.86e+00		1.39e+00		**0.00e+00**		6.26e-01		1.95e+00
F5	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F6	Mean	6.83e-02	=	4.71e-01	+	9.22e-02	=	**2.46e-02**	-	1.22e-01	=	1.82e-01	=	5.91e-02
	Std	4.76e-02		8.80e-02		7.46e-02		**2.43e-02**		1.07e-01		4.34e-01		4.14e-02
F7	Mean	**0.00e+00**	=	6.24e-01	=	**0.00e+00**	=	2.28e-09	=	6.24e-02	=	1.19e-05	=	**0.00e+00**
	Std	**0.00e+00**		1.14e-01		**0.00e+00**		1.00e-08		1.90e-01		2.43e-05		**0.00e+00**
F8	Mean	1.51e-02	+	7.76e-02	+	7.88e-03	=	3.17e-02	+	4.51e-02	+	5.67e-02	+	**4.21e-03**
	Std	1.74e-02		2.55e-02		9.56e-03		3.69e-02		4.89e-02		3.60e-02		**4.85e-03**
F9	Mean	2.17e+02	=	**1.86e+02**	-	2.22e+02	=	2.29e+02	=	2.29e+02	=	2.29e+02	=	2.13e+02
	Std	4.73e+01		**0.00e+00**		4.19e+01		0.00e+00		0.00e+00		0.00e+00		5.82e+01
F10	Mean	6.40e+01	+	2.00e+02	+	1.42e+01	+	1.00e+02	+	1.00e+02	+	1.00e+02	+	**0.00e+00**
	Std	4.03e+01		2.53e+01		2.94e+01		2.28e-02		2.49e-02		2.95e-02		**0.00e+00**
F11	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F12	Mean	**1.59e+02**	-	1.59e+02	-	1.59e+02	-	1.59e+02	-	1.65e+02	+	1.64e+02	+	1.60e+02
	Std	**3.68e-01**		4.21e+00		8.30e-01		1.37e-01		4.70e-01		1.04e+00		1.35e+00
	+/=/-	2+/8=/2-		4+/5=/3-		1+/8=/1-		2+/7=/3-		4+/7=/1-		3+/8=/1-		
							20D							
F1	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F2	Mean	**0.00e+00**	=	3.99e+00	=	4.28e+01	+	4.91e+01	+	4.22e+01	+	4.91e+01	+	1.34e-01
	Std	**0.00e+00**		1.51e+00		1.42e+01		0.00e+00		1.45e+01		0.00e+00		7.36e-01
F3	Mean	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**	=	**0.00e+00**
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**
F4	Mean	1.98e+01	-	2.09e+01	-	4.65e+01	-	2.38e+01	-	**0.00e+00**	-	4.08e+00	-	5.80e+01
	Std	2.64e+00		4.03e+00		7.16e+00		4.37e+00		**0.00e+00**		5.45e-01		7.83e+00
F5	Mean	**0.00e+00**	-	**0.00e+00**	-	**0.00e+00**	-	**0.00e+00**	-	**0.00e+00**	-	**0.00e+00**	-	3.89e+02
	Std	**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		**0.00e+00**		1.39e+02
F6	Mean	2.61e-01	=	1.06e+00	+	2.87e-01	+	4.88e-01	+	4.65e-01	+	3.33e+01	+	**2.52e-01**
	Std	8.02e-02		2.14e-01		9.12e-02		1.86e-02		2.53e-02		5.30e+00		**1.29e-01**
F7	Mean	1.87e+00	+	2.03e+01	=	1.87e+00	+	2.04e-01	-	**1.35e-01**	+	2.35e+00	+	1.49e+00
	Std	9.82e-01		3.57e+00		7.60e-01		3.16e-01		**1.77e-01**		1.30e+00		1.87e+00
F8	Mean	1.28e+01	=	2.07e+01	=	1.74e+01	=	1.83e+01	-	**3.06e+00**	-	1.99e+01	+	1.74e+01
	Std	7.86e+00		8.58e+00		2.75e+00		5.02e+00		**6.86e+00**		1.05e+00		4.52e+00
F9	Mean	1.81e+02	=	**1.65e+02**	-	1.81e+02	=	1.81e+02	=	1.81e+02	=	1.81e+02	=	1.81e+02
	Std	0.00e+00		**0.00e+00**		0.00e+00		0.00e+00		0.00e+00		1.90e-07		0.00e+00
F10	Mean	9.77e+01	+	1.00e+02	+	**0.00e+00**	=	1.00e+02	+	1.00e+02	+	1.00e+02	+	**0.00e+00**
	Std	1.06e+01		3.55e-02		**0.00e+00**		2.22e-02		1.31e-02		1.60e-02		**0.00e+00**
F11	Mean	5.00e+01	+	4.00e+02	+	3.00e+02	+	3.00e+02	+	2.97e+02	+	3.00e+02	+	**0.00e+00**
	Std	1.14e+02		4.98e+01		0.00e+00		0.00e+00		8.90e+01		0.00e+00		**0.00e+00**
F12	Mean	2.31e+02	-	**2.00e+02**	-	2.34e+02	=	2.29e+02	-	2.33e+02	=	2.34e+02	=	2.34e+02
	Std	1.75e+00		**1.92e+00**		1.62e+00		1.19e+00		1.76e+00		4.54e+00		1.76e+00
	+/=/-	3+/5=/4-		3+/5=/4-		4+/6=/2-		4+/4=/4-		4+/4=/4-		6+/4=/2-		

Table S8: Statistical Result of Ablation Study. w/o-LDS: a variant where the linear dynamic strategy is removed. w/o-FCS: a variant where the "Fixed Cluster" strategy is entirely removed.

Algorithm	Wilcoxon Rank	Friedman Rank
CAMODE-CLA	–	**1.58**
w/o-LDS	57+/18=/9-	2.45
w/o-FCS	68+/12=/4-	2.97

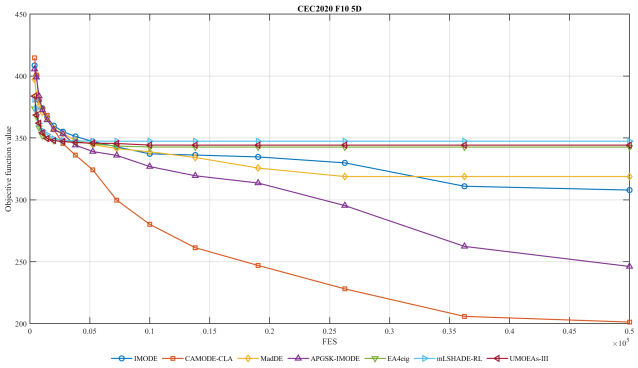

Figure S1: The convergence graph on CEC2020 F10 5D

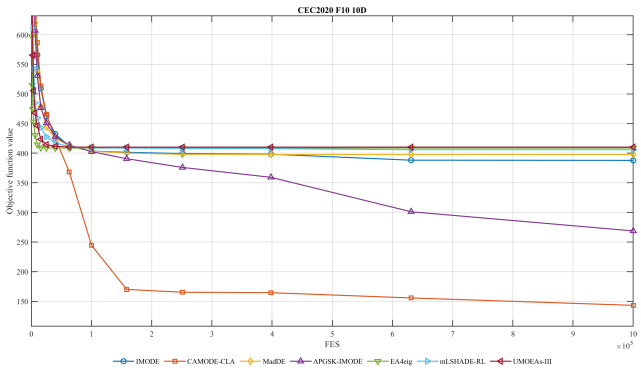

Figure S2: The convergence graph on CEC2020 F10 10D

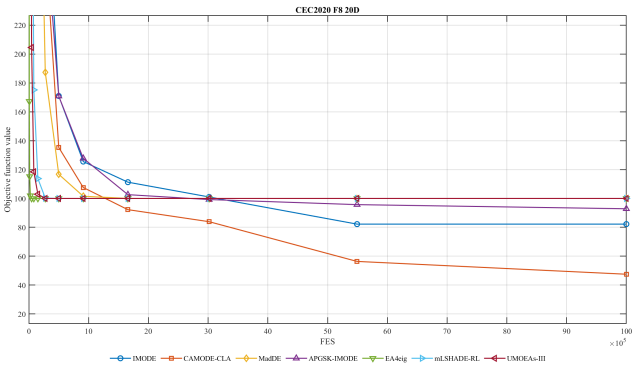

Figure S3: The convergence graph on CEC2020 F8 20D

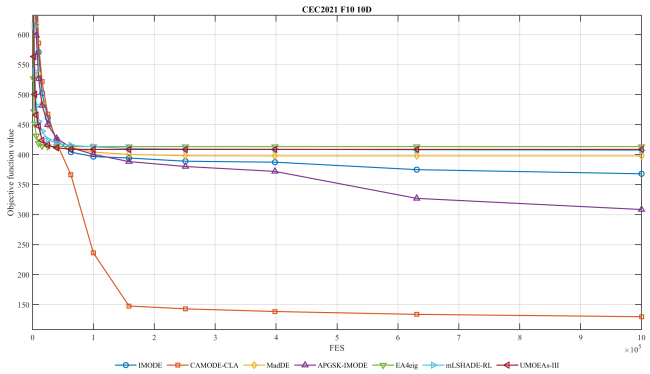

Figure S4: The convergence graph on CEC2021 F10 10D

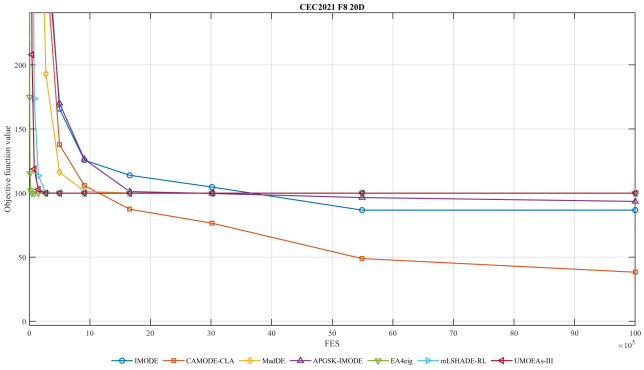

Figure S5: The convergence graph on CEC2021 F8 20D

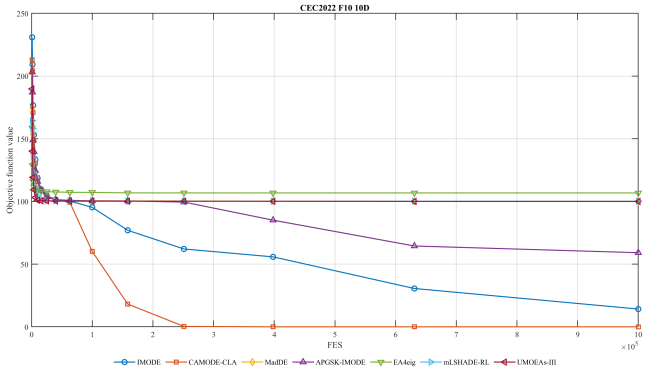

Figure S6: The convergence graph on CEC2022 F10 10D